# Lahja at DialectSentEval Shared Task Subtask-1: Seed-Ensembled MARBERT for Multi-Dialect Arabic Sentiment Analysis with Zero-Shot Lebanese Transfer

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## Abstract

Arabic sentiment analysis must cope with dialects that differ in vocabulary, morphology and spelling. Subtask 1 of the DialectSentEval 2026 shared task poses this problem directly: systems label hotel review sentences from five Arabic dialects as positive, negative or neutral, and one dialect (Lebanese) appears only in the test set. In this paper, we describe the Lahja submission. Since the training corpus is small (1,731 sentences) and consists of parallel translations, we treat unreliable measurement (i.e. leaky validation splits and single training runs) rather than model capacity as the main risk. We fine-tune MARBERTv2 under three random seeds, average their softmax outputs, and validate on a split grouped by review identifier so that all dialect versions of a review stay on the same side. The ensemble reached 0.790 macro-F1 on the official test set, ranking 10th of 12 teams, and lost only about two points from validation to test despite the unseen dialect. Our ablations show that the variation between seeds (about 1.5 points) exceeds the effect of orthographic normalisation or class weighting, and that the normalised ensemble, although best on validation, generalised worst on test. We therefore recommend grouped validation and seed-spread reporting as standard practice for small-data shared tasks.

## 1 Introduction

Sentiment analysis for Arabic is complicated by the dialectal diversity of the language. The varieties spoken across the Arab world differ in vocabulary, morphology and orthographic habits, and systems trained on one variety often degrade on others. The DialectSentEval 2026 shared task (Ezzini et al., 2026a) is the successor to the AraSentEval pilot (Ezzini et al., 2026b) and includes a companion sentiment-swapping subtask built on the MA'AKS dataset (Mughaus et al., 2026). Subtask 1 frames the dialect challenge concretely: systems classify hotel review sentences from five dialects as positive, negative or neutral, and are evaluated by macro-F1. The twist that shaped our approach is that one dialect, Lebanese, appears *only* in the test set, so a fifth of the evaluation is zero-shot dialect transfer.

With fewer than two thousand training sentences arranged as parallel translations, the question we set out to answer was a practical one: can such a corpus support trustworthy system development at all? Anyone who has tuned a classifier on a corpus of this size knows how easily a lucky seed can pass for an improvement. Therefore, we judged that the main risk was not underfitting but *mismeasurement* (i.e. validation splits that leak parallel content and conclusions drawn from single runs), and we wrapped a deliberately simple system, a fine-tuned MARBERTv2 ensembled over three seeds, in careful measurement.[1] The salient contributions of this paper are as follows. First, we present a seed-ensembled MARBERTv2 system that reached 0.790 test macro-F1 (10th of 12 final-phase teams) and was trained in under ten minutes on a consumer laptop. Second, we describe an ID-grouped validation protocol for parallel dialect corpora whose dev scores predicted official test performance within 1.5 to 2.3 points despite the unseen dialect. Third, we report a transparent ablation study showing that seed variance (about 1.5 F1) exceeds the effect of orthographic normalisation and class weighting, neither of which transferred to the test set.

## 2 Related Work

Arabic sentiment analysis has largely moved from feature-based models over Modern Standard Arabic (MSA) to pretrained transformers. Dialect-focused encoders, such as MARBERT (Abdul-

[1] Leaderboard account: shadenalzeer.



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Mageed et al., 2021), trained on tweet-scale dialectal text, and the CAMeLBERT family (Inoue et al., 2021), which studies the interplay of language variant and task, consistently outperform MSA-centric models such as AraBERT (Antoun et al., 2020) on dialectal benchmarks. Shared tasks have driven much of this progress. AHaSIS (Alharbi et al., 2025a) and the related ADOR benchmark (Alharbi et al., 2025b) evaluated hotel review sentiment in Saudi and Moroccan dialects (twelve systems, best macro-F1 0.81), large language models have been evaluated on the same problem (Alharbi et al., 2025c), and DialectSentEval builds on the AraSentEval pilot (Ezzini et al., 2026b). To our knowledge, it is among the first shared tasks to evaluate sentiment transfer to a dialect that is entirely unseen in training. Our contribution is less a new architecture than a measurement protocol for such parallel, low-resource settings, following work that shows that seed variance in fine-tuning rivals reported improvements (Reimers and Gurevych, 2017; Dodge et al., 2020; Mosbach et al., 2021) and that the way a corpus is split shapes the conclusions drawn from it (Gorman and Bedrick, 2019; Søgaard et al., 2021).

## 3 Data

The Subtask 1 release of the MDS-3 corpus (Ezzini et al., 2026a) contains 1,731 training sentences in four dialects (Moroccan Darija, Egyptian, Jordanian and Saudi, approximately 433 each) and 525 blind test sentences in five dialects (105 each), with **Lebanese appearing only at test time**. The sentences are hotel reviews rendered into each dialect by native speakers. Therefore, the corpus has a *parallel* structure: the same underlying review appears in every dialect under a shared identifier (ID). Training has 376 review IDs; reviews of more than one sentence share an ID, and one review lacks an Egyptian version, hence 1,731 sentences. The labels are *positive*, *negative* and *neutral* (38%, 35% and 27% of the training data, respectively). Interestingly, 201 of the 376 training review IDs carry *different* gold labels across their dialect versions, which suggests that dialectal rendering can shift perceived sentiment.

To gauge the difficulty of the unseen dialect, we measured word-level Jaccard overlap between dialect vocabularies (Figure 1). Training dialects overlap each other at 0.19 to 0.48, whereas the Lebanese test vocabulary overlaps each training dialect at only 0.07 to 0.11. Zero-shot transfer to Lebanese therefore cannot rely on shared surface vocabulary and must come from the pretrained encoder. For transparency, we note that 58 Egyptian test sentences occur verbatim, with labels, in the training data (an artefact of the data preparation); we did not exploit this overlap in any way.

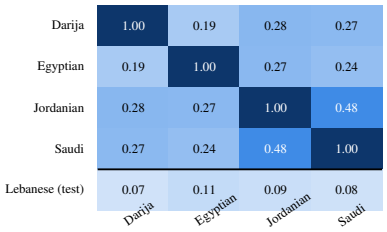



Figure 1: Word-level Jaccard overlap between dialect vocabularies. Training dialects (above the line) share 0.19 to 0.48 of their vocabulary, whereas the test-only Lebanese data (below the line) share just 0.07 to 0.11 with every training dialect. Therefore, zero-shot transfer cannot rely on lexical memorisation.

## 4 System Description

**Validation protocol.** As the corpus is parallel, a conventional row-level random split would place different dialect versions of the *same* review on both sides of the train/validation boundary, and a model could then score on content that it has effectively seen. We instead split *by review ID* (85/15, stratified by the most frequent label of each ID, with a fixed seed), keeping all dialect versions of a review on the same side. This yields 1,467 training and 264 validation sentences with zero content overlap. As Section 5 shows, this protocol predicted official test performance within 1.5 to 2.3 macro-F1 points although 20% of the test set is an unseen dialect, and we therefore consider it a prerequisite for trustworthy development on parallel dialect corpora.

**Model.** Our backbone is MARBERTv2 (Abdul-Mageed et al., 2021), a BERT-base encoder pretrained on roughly one billion Arabic tweets and further pretrained on MSA text with 512-token sequences. The choice follows directly from the vocabulary analysis in Section 3. Since zero-shot Lebanese transfer cannot rely on surface overlap with the training dialects, it must come from the pretraining of the encoder, and tweet-scale pretraining offers the broadest dialectal exposure available. We fine-tune the encoder with a standard classification head using the Transformers li-

| System | Dev F1 | Test F1 |
|---|---|---|
| TF-IDF + SVM | 0.728 | 0.709† |
| MARBERTv2 (seed 42) | 0.802 | 0.787 |
| MARBERTv2 (3 seeds) | 0.802±0.008 | — |
| + class-weighted loss | 0.792 | — |
| + normalisation (3 seeds) | 0.803±0.024 | — |
| Normalised 3-seed ensemble | 0.814 | 0.768 |
| **3-seed ensemble (final)** | **0.813** | **0.790** |

Table 1: Macro-F1 on the internal dev split and the official test set; ± rows give the mean and standard deviation over seeds 42, 43 and 44; "—" marks configurations not submitted; †classical-track submission.

brary (Wolf et al., 2020), with a learning rate of $2\times10^{-5}$, a batch size of 16, 4 epochs, AdamW with 10% linear warm-up, a weight decay of 0.01 and a maximum sequence length of 128 (the 99th percentile of MARBERT token lengths in the data is 77, so no truncation occurs in practice). After each epoch we evaluate on the validation split and keep the checkpoint with the best macro-F1 (full configuration in Appendix A).

**Final system.** We train three runs that differ only in their random seed (42, 43 and 44) and average their softmax distributions at inference, which is the deep-ensemble recipe of Lakshminarayanan et al. (2017); majority voting can tie with three models, whereas probability averaging cannot.[2] Each run takes about three minutes on a consumer laptop (Apple M-series, MPS backend), so the entire final system costs under ten minutes of compute. As a sanity floor, we also report a linear SVM over TF-IDF features (word 1–2-grams and character 3–5-grams).

## 5 Experiments and Results

Table 1 summarises our experiments. The classical floor (TF-IDF features with a linear SVM) reaches 0.728 dev and 0.709 test macro-F1. A single fine-tuned MARBERTv2 adds roughly seven points (0.802 dev and 0.787 test), and averaging three seeds adds a further 0.3 (test) to 1.1 (dev) points. This yields our final official score of **0.790 test macro-F1** (accuracy 0.813), which placed our system **10th of the 12 teams** on the final-phase leaderboard. However, ranks 5 to 11 span only 1.5 macro-F1 points (0.801 to 0.786), a band no wider than the seed variance we measure in Section 6. Therefore, mid-table orderings in this task

[2]Code: https://github.com/Shaden-Saad/DialectSentEval2026.

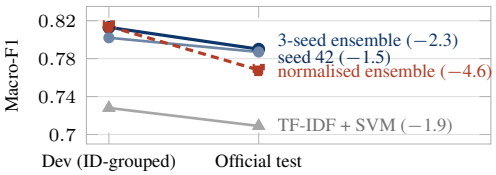



Figure 2: Dev-to-test generalisation of the four submitted systems (Table 1): raw-text systems lose 1.5 to 2.3 points, whereas the normalised ensemble loses 4.6.

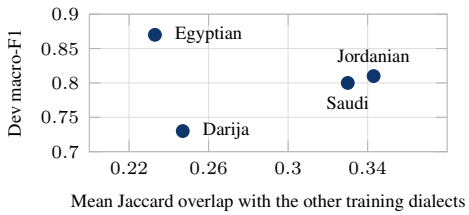



Figure 3: Per-dialect dev macro-F1 (single seed) against lexical distance (mean Jaccard overlap in Figure 1): the two most distinct dialects are the easiest and the hardest.

are likely to be statistically fragile (Dror et al., 2018).

Two comparisons are as informative as the headline number. First, Figure 2 plots the *dev-to-test gap* of each submitted system: our raw-text submissions lost only 1.5 to 2.3 points from dev to test, even though one fifth of the test set is a dialect absent from training, which is evidence both that ID-grouped validation is trustworthy and that the dialectal pretraining of MARBERT transfers to an unseen dialect. By contrast, the normalised ensemble scored highest on dev (0.814) yet dropped 4.6 points on test (0.768), the largest generalisation failure we observed; Section 6 examines why. Second, Figure 3 plots per-dialect dev scores against the vocabulary distances of Section 3, and the two only partially agree: the most lexically distinct dialects sit at opposite ends of the range, as Darija is hardest (0.73 macro-F1) while Egyptian is easiest (0.87), with Jordanian and Saudi in between (0.81 and 0.80; Appendix B).

## 6 Analysis

**Where the errors live.** Of the 47 errors that our single-seed model makes on the validation split, 45 involve the *neutral* class; the positive/negative boundary is almost solved (two errors). Typical failures are mixed-evaluation sentences, such as the Darija review *shi ghuraf 3andhum mandhar, u shi ghuraf ma 3andhumsh* ("some rooms have

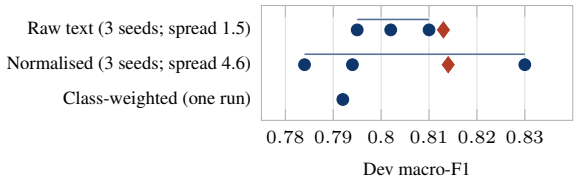



Figure 4: Seed variance versus interventions on dev: single runs (dots) span 1.5 points for raw text, normalisation widens the spread to 4.6, and the two ensembles (diamonds) are indistinguishable.

a view and some do not"; gold NEUTRAL, predicted NEGATIVE), and pleasant-topic descriptive sentences (e.g. "one minute from the nightclubs, cafés and restaurants"; gold NEUTRAL, predicted POSITIVE), where the gold label itself is sometimes debatable. Error position is also dialect-sensitive: of the 27 reviews with at least one erroneous dialect version, 16 are misclassified in *exactly one* dialect while their parallel versions are handled correctly, so most errors are triggered by the surface realisation of a review rather than by its content.

**Seed variance exceeds every intervention we tried.** Figure 4 shows every run we trained. Across seeds 42, 43 and 44, the identical configuration scores 0.802, 0.810 and 0.795 dev macro-F1, a spread of 1.5 points, and no preprocessing intervention we evaluated had a larger effect. Inverse-frequency class weighting *lost* a net 1.0 point (Appendix B), and orthographic normalisation (i.e. diacritic stripping, alef/ya/teh-marbuta unification and de-elongation) moved individual seeds by $-1.8$, $+2.0$ and $-0.1$ points, which is indistinguishable from noise. We conclude that single-run comparisons on corpora of this size are not interpretable, and we encourage future shared-task reports to publish seed spreads alongside headline scores (Reimers and Gurevych, 2017).

**Normalisation fails exactly where it must not.** Although the ensemble of three *normalised* runs scored highest on dev (0.814), it collapsed on test (0.768), a 4.6-point generalisation gap that is roughly double the largest gap among our raw-text submissions. Our hypothesis is that normalisation deletes precisely the orthographic cues that distinguish dialects, so MARBERT, which is pretrained on raw, un-normalised tweets, receives Lebanese input unlike anything that normalisation produced from the training dialects. Without per-dialect test scores we cannot verify this directly. However, a companion classical system that we built with comparable orthographic normalisation (a TF-IDF soft-voting ensemble of logistic regression and complement naive Bayes) showed the same signature, dropping from 0.763 under ID-grouped cross-validation to 0.668 on test. Finally, a six-model ensemble mixing raw and normalised inputs underperformed both three-model trios (0.810 dev), which suggests that averaging probabilities across models that read *different representations* of the input dilutes rather than diversifies the signal.

## 7 Conclusion

In this paper, we have described a simple, reproducible and rigorously measured system (MARBERTv2, three seeds and softmax averaging) that proved competitive on multi-dialect Arabic sentiment analysis and transferred to an unseen dialect with a dev-to-test gap of only two points. The interventions that promised more (class weighting and normalisation) delivered less, and the one that looked best on dev generalised worst on test. Our broader takeaway for small-data shared tasks is methodological: grouped validation and seed-spread reporting are cheap, and without them both system development and leaderboard interpretation rest on noise. Future work includes dialect-adaptive pretraining and better modelling of the neutral class.

## Limitations

The training set is small (1,731 sentences) and single-domain, so our conclusions may not transfer to other genres. Without per-dialect test scores, our hypothesis that normalisation harms zero-shot Lebanese remains indirect. Three seeds estimate variance only coarsely, and some neutral-class errors reflect ambiguous gold labels (201 of 376 review IDs change label across dialects).

## Ethical Considerations

We use the MDS-3 data solely under the terms of the shared task, do not redistribute them, and disclosed rather than exploited the train/test overlap (Section 3). Our per-dialect gaps (14 dev F1 points between Darija and Egyptian) show that deployment without per-dialect evaluation could disadvantage speakers of less-resourced varieties.

## Acknowledgements

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## Appendices

The appendices record the configuration and the per-run results behind Sections 4 to 6 so that the system can be reproduced. They contain no new experiments: every number either appears in the main text or is derived from numbers that do.

## A Training and Evaluation Details

Table 2 lists the configuration of the final system, of the two ablations and of the classical systems.

## B Per-Run and Per-Dialect Results

Table 3 gives the dev macro-F1 of every run behind Table 1 and Figure 4. The per-seed scores of the normalised runs are derived from the raw-text scores and the per-seed changes reported in Section 6. Class weighting raised neutral recall from 0.66 to 0.70 but sacrificed neutral precision and the positive/negative boundary, hence its net loss of 1.0 point. Table 4 gives the per-dialect dev scores of the single-seed model together with the lexical distance used in Figure 3.

| Dialect | Dev macro-F1 | Mean Jaccard overlap |
|---|---|---|
| Moroccan Darija | 0.73 | 0.247 |
| Egyptian | 0.87 | 0.233 |
| Jordanian | 0.81 | 0.343 |
| Saudi | 0.80 | 0.330 |

Table 4: Per-dialect dev macro-F1 of the single-seed model and the mean word-level Jaccard overlap of each dialect with the other three training dialects (from Figure 1).

| Component | Setting |
|---|---|
| Data split | By review ID, 85/15, stratified by the most frequent label of each ID, fixed seed; 1,467 training and 264 validation sentences |
| Backbone | MARBERTv2 (BERT-base; tweet pretraining, then MSA continued pretraining with 512-token sequences) |
| Head | Standard sequence-classification head, three classes |
| Optimiser | AdamW, learning rate $2\times10^{-5}$, weight decay 0.01, 10% linear warm-up |
| Batch size, epochs | 16, 4 |
| Sequence length | 128 tokens (99th percentile of token lengths in the data: 77) |
| Checkpoint | Best validation macro-F1 across epochs |
| Seeds | 42, 43, 44 (otherwise identical configuration) |
| Ensembling | Mean of the softmax distributions of the three runs at inference |
| Hardware, time | Apple M-series laptop (MPS backend); about three minutes per run, under ten minutes in total |
| Software | Transformers library (Wolf et al., 2020) |
| Normalisation (ablation) | Diacritic stripping, alef/ya/teh-marbuta unification and de-elongation applied to the input text |
| Class weighting (ablation) | Inverse-frequency class weights (computed on the training split, normalised to mean 1) in the cross-entropy loss |
| Classical baseline | Linear SVM (balanced class weights) over TF-IDF word 1–2-grams and character 3–5-grams (classical-track submission) |
| Companion classical system | TF-IDF soft-voting ensemble of logistic regression and complement naive Bayes on normalised input; evaluated with ID-grouped cross-validation |

Table 2: Configuration of the systems reported in this paper.

| | Dev, per seed | | | Ensemble | |
|---|---|---|---|---|---|
| Configuration | 42 | 43 | 44 | Dev | Test |
| Raw text | 0.802 | 0.810 | 0.795 | 0.813 | 0.790 |
| Normalised | 0.784 | 0.830 | 0.794 | 0.814 | 0.768 |
| Class-weighted | 0.792 (one run) | | | — | — |
| Raw + normalised | six models above | | | 0.810 | — |

Table 3: Dev macro-F1 of the individual runs and of the ensembles. Raw-text runs have a mean of 0.802 and a standard deviation of 0.008; normalised runs 0.803 and 0.024.